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Empirical Article

Clinical Psychological Science
2018, Vol. 6(3) 372–381
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DOI: 10.1177/2167702617748401
www.psychologicalscience.org/CPS

Adaptation to Vocal Expressions and Phonemes Is Intact in Autism Spectrum Disorder

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Abstract

Several recent studies have demonstrated reduced visual aftereffects, particularly to social stimuli, in autism spectrum disorder (ASD). This putative impairment of the adaptive mechanism in ASD has been put forward as a possible explanation for some of the core social problems experienced by children with ASD (e.g., facial emotion or identity recognition). We addressed this claim in children with ASD and typically developing children by using an established methodology and morphed auditory stimulus set for eliciting robust aftereffects to vocal expressions and phonemes. Although children with ASD were significantly worse at categorizing the vocal expressions compared with the control stimuli (phoneme categorization), aftereffect sizes in both tasks were identical in the two participant groups. Our finding suggests that the adaptation mechanism is not universally impaired in ASD and is therefore not an explanation for the social perception difficulties in ASD.

Keywords

autism, adaptation, aftereffects, voice, emotion

Received 4/12/17; Revision accepted 10/27/17

Introduction

Autism spectrum disorder (ASD) is a heritable, neurodevelopmental condition, which is in part characterized by lifelong impairments in social communication and interaction (American Psychiatric Association, 2013) as well as altered sensory processing (Marco, Hinkley, Hill, & Nagarajan, 2011). Difficulties with processing the social information from faces and voices are a commonly cited aspect of ASD, particularly with regard to identifying and discriminating between facial identities (see Weigelt, Koldewyn, & Kanwisher, 2012, for a review) and emotional expressions from both faces (see Harms, Martin, & Wallace, 2010, for a review; Uljarevic & Hamilton, 2013) and voices (e.g., Globerson, Amir, Kishon-Rabin, & Golan, 2015; Hobson, Ouston, & Lee, 1988; Stewart, McAdam, Ota, Peppe, & Cleland, 2013). Pellicano, Jeffery, Burr, and Rhodes (2007) have suggested that one underlying cause of face perception difficulties may be an impaired adaptation mechanism.

Adaptation is thought to be a universal mechanism across all sensory systems (Møller, 2002) and refers to a change in neural firing and, therefore, behavior as a result of continued stimulation or exposure to the same sensory stimulus (Quiroga & Panzeri, 2013). The generally accepted view is that adaptation is essential for efficient coding and processing of stimuli by allowing novel features to appear more salient (Webster, 2011). Research using adaptation has revealed neural populations tuned

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to respond to specific stimulus attributes by isolating and subsequently distorting the perception of these attributes (Grill-Spector et al., 1999; Winston, Henson, Fine-Goulden, & Dolan, 2004). Adaptation has therefore been a useful tool to investigate different levels of processing, including low-level stimulus properties such as motion, shape, or color (e.g., Wright, 1934) and high-level properties such as facial or vocal features (e.g., emotional expressions).

Webster and MacLin (1999) were the first to show that extended exposure to faces can generate aftereffects, that is, a perceptual bias in which the test face was interpreted as being distorted in the opposite direction of that of the adaptor face. Adaptation to consistently distorted faces (e.g., expanded features) caused subsequently viewed unmanipulated faces to appear distorted in the opposite direction of the adapting stimulus (e.g., compressed features). This effect transferred to faces of different identities. Later studies demonstrated that these contrastive aftereffects are robust to changes in the retinal position (Leopold, O'Toole, Vetter, & Blanz, 2001), size (Zhao & Chubb, 2001), and angular orientation (Watson & Clifford, 2003) of the adapting stimulus, suggesting that adaptation to those low-level features cannot fully explain face aftereffects. Thus, adaptive coding is a strategy employed by the neural system throughout the perceptual processing hierarchy and is not only employed at early stages of visual information processing. Moreover, even high-level, socially important properties such as facial identity (Leopold et al., 2001), attractiveness (Rhodes, Jeffery, Watson, Clifford, & Nakayama, 2003), eye gaze (Jenkins, Beaver, & Calder, 2006), and expression (Fox & Barton, 2007; Webster, Kaping, Mizokami, & Duhamel, 2004) can be isolated using adaptation. Researchers have interpreted the resulting aftereffects as reflecting a neural strategy for optimizing perception, including a recalibration of neural processes to continuously updated stimulation (Webster & MacLin, 1999). Whereas research has used these contrastive aftereffects as a tool to evaluate face perception models (e.g., Bestelmeyer et al., 2008; Bestelmeyer, Jones, DeBruine, Little, & Welling, 2010; Jones, Feinberg, Bestelmeyer, DeBruine, & Little, 2010; Leopold et al., 2001; Leopold, Rhodes, Muller, & Jeffery, 2005; Little, DeBruine, & Jones, 2005; Webster et al., 2004; Welling et al., 2009), it has recently been shown that corresponding auditory aftereffects can also be elicited for voices (e.g., identity: Zäske, Schweinberger, & Kawahara, 2010; gender: Schweinberger et al., 2008; age: Zäske & Schweinberger, 2011; and affect: Bestelmeyer, Rouger, DeBruine, & Belin, 2010; Skuk & Schweinberger, 2013).

In an influential article, Pellicano et al. (2007) reported that children with ASD show significantly reduced facial identity aftereffects compared with IQ-matched, typically developing children. Although face identification was actually unimpaired in both groups, the authors suggested that an impaired adaptive face-coding mechanism in ASD may be responsible for the commonly reported face perception problems as well as some of the core social deficits in ASD. A follow-up study ascertained that these reduced aftereffects in ASD are not caused by a lack of attention to the adapting stimulus (Ewing, Leach, Pellicano, Jeffery, & Rhodes, 2013), supporting that children with ASD have difficulties with adaptive norm-based coding of face identity. Siblings and parents of children with ASD also show this reduction in aftereffect size to identity adaptation, suggesting that this behavioral difference may have the potential to serve as an endophenotype for ASD (Fiorentini, Gray, Rhodes, Jeffery, & Pellicano, 2012). Recent evidence of reduced adaptation despite intact recognition of biological motion (van Boxtel, Dapretto, & Lu, 2016), another social stimulus, further strengthens the possibility that changes in the adaptation mechanism contribute to the social phenotype in ASD.

The literature is not entirely unequivocal regarding reduced aftereffects to social stimuli in ASD. For example, recent studies have shown that reduced aftereffects to nonsocial stimuli can also be observed, for example, reduced aftereffects to numerosity (Turi et al., 2015) and audio-visual asynchrony (Turi, Karaminis, Pellicano, & Burr, 2016), although not consistently to all nonsocial stimuli, for example, intact adaptation to perceptual causality (Karaminis et al., 2015). Similarly, it is interesting that many of the reduced face aftereffects found in children with ASD cannot be replicated in adults with ASD (e.g., Cook, Brewer, Shah, & Bird, 2014; Walsh, Maurer, et al., 2015; Walsh, Vida, Morrisey, & Rutherford, 2015). In isolation, this last pattern of results would suggest that reduced contrastive aftereffects for faces in children with ASD reflect delayed or atypical development of adaptive mechanisms and, as such, may not be a core feature of the disorder. However, this notion is in contrast with Fiorentini et al.'s (2012) finding of reduced visual aftereffects in relatives (including adult relatives) of ASD children. If future work confirms that reduced face aftereffects are merely due to developmental delays, they are unlikely to be specific to autism and may not be useful as an endophenotype of ASD.

In light of these findings, several questions remain unanswered regarding the specificity of reduced or sometimes even absent contrastive aftereffects in ASD. First, it is unclear from the current literature whether this alteration in adaptation strength is specific to the stimulus type (e.g., social or nonsocial), developmental stage, or modality. Only one article has explored adaptation in ASD in the auditory domain. Lawson, Aylward, White, and Rees (2015) showed that continuous

stimulation with sound of a certain volume caused a reduced magnitude and rate of simple loudness adaptation in children with ASD but not in typically developing children. Although this finding theoretically provides support for a generalized reduction in adaptation in ASD, it may instead be in line with increased sound sensitivity in ASD and does not rule out the possibility that adaptive mechanisms to more complex sound categories (e.g., social stimuli) are intact. Thus, whether children with ASD also show reduced *contrastive* aftereffects in response to adaptation using more complex auditory stimuli has not been determined. The advantage of using auditory over visual stimuli to assess the specificity of the reduced aftereffect to certain stimulus types (social vs. nonsocial) is that even distractible children cannot ignore auditory stimuli presented via headphones as easily as visually presented stimuli on a small computer display.

The aim of the current study was to address some of these remaining questions. First, we examined whether reduced contrastive aftereffects in ASD also extend to high-level stimuli in the auditory domain. Second, we assessed whether aftereffects are smaller or even absent in a task with emotionally salient stimuli compared with a control task in ASD. More specifically, we were interested in group differences in the perceptual plasticity of two vocal features with differing emotional content, which could form part of the phenotype of ASD. Using auditory voice morphing, we created blends between two instances of two stimulus categories: emotional vocalizations (anger and fear) and phonemes (/m/ and /o/). Given the previous literature, we predicted both a main effect of participant group and a significant interaction between stimulus type and group when measuring the magnitudes of contrastive aftereffects. In other words, we expected a reduction in aftereffects across all stimuli in the ASD group as well as a greater aftereffect reduction in ASD to the more socially salient emotion stimuli, compared with the phoneme stimuli. We expected the typically developing children to show robust aftereffects across condition and be unaffected by stimulus type.

Methods

Participants

On the basis of the between-group difference reported by Pellicano et al. (2007), we performed a power analysis in G*Power 3 (Faul, Erdfelder, Buchner, & Lang, 2009). This analysis determined that at an alpha level of .05 (two-tailed), an effect size of $d = .81$, and a power of $\beta = .8$, a sample size of 25 participants is required in each group. We therefore recruited a total of 30 Caucasian children with ASD as outpatients of the Department of Child and Adolescent Psychiatry, Psychotherapy, and Psychosomatics, Medical Faculty, University of Freiburg, along with 26 age- and culture-matched, typically developing controls. As a result of incomplete data sets or misunderstandings of the task, we had to exclude 3 children with ASD and 2 typically developing children. Our analyses are therefore based on 27 (2 female) children with ASD and 24 (8 female) typically developing children. Table 1 summarizes age and IQ as measured by the Cattell Culture Fair Intelligence Test (Weiss, 2006) for the included participants of both groups except for 1 of the children with ASD, for whom IQ was assessed using the Snijders-Oomen nonverbal IQ test (Tellegen, Winkel, Wijnberg-Williams, & Laros, 1998). His score was 78 and is not included in Table 1. Children with ASD had a diagnosis according to criteria of the ICD-10 and the fourth edition of the *Diagnostic*

Table 1. Participant Demographics and Test Scores

	Children with autism				Typically developing children					
Measure	*N*	*M*	*SD*	Range	*N*	*M*	*SD*	Range	*t*	*p*
Age	27	11.87	2.38	8.10–17.69	24	12.20	2.14	6.46–15.22	–0.52	.607
IQ (CFT 20-R)	25	96.00	14.02	71–122	24	109.08	14.89	84–141	–3.26	.002
SRS raw scores	27	99.48	33.82	30–170						
SRS T-scores	27	80.89	11.79	55–100						
ADOS (social)	27	7.85	1.83	4–11						
ADOS (communication)	27	3.89	2.04	0–8						
ADOS (total)	27	11.74	3.05	6–17						
ADI-R (social)	26	18.58	5.14	10–26						
ADI-R (communication)	26	12.77	5.18	1–21						
ADI-R (repetitive behavior)	26	5.12	3.34	0–13						

Note. CFT 20-R = Cattell Culture Fair Intelligence Test; SRS = Social Responsiveness Scale; ADOS = Autism Diagnostic Observation Schedule; ADI-R = Autism Diagnostic Interview.

and Statistical Manual of Mental Disorders (*DSM–IV*; American Psychiatric Association, 1994). Diagnoses of ASD are based on the international standard, including the German versions of the Autism Diagnostic Observation Schedule (ADOS; Rühl, Bölte, Feineis-Matthews, & Poustka, 2004), the Social Responsiveness Scale (SRS; Bölte & Poustka, 2007), and the Autism Diagnostic Interview (ADI-R; Bölte, Rühl, Schmötzer, & Poustka, 2006), which is a semistructured interview of caregivers and their observations of their child. Table 1 provides means, standard deviations, and ranges of the social, communication, and total ADOS score; the SRS raw scores and T-scores; and the three categories indicating autistic behavior of the ADI-R ("social interaction," "social communication," and "restricted, repetitive, stereotyped behaviors") for the children with ASD. In our sample, all participants but 1 were above the cutoff in the ADI-R category "social interaction," only 1 was under the cutoff for "communication," and 4 participants did not reach the cutoff for "restricted, repetitive, stereotyped behaviors." Overall, all patients exceeded the cutoff in at least two domains of the ADI-R (which is a criterion for a diagnosis of ASD) and also scored in the third "autistic" domain (although often under the cutoff). Moreover, in the ADOS, all patients but 2 (who scored slightly under the cutoff) received an ADOS diagnosis of autism spectrum or autism. These 2 patients were above the cutoff in at least two domains of the ADI-R. Only 1 participant (not previously mentioned) scored under the cutoff for deficits of social responsiveness.

Our sample is high functioning with good cognitive and communication abilities, and the severity of symptoms can be considered at level 2 (middle) according to the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM–5*; American Psychiatric Association, 2013). The symptom severity of our sample is comparable to that of Pellicano et al. (2007). This study was approved by the ethics committee of Bangor University for piloting with healthy adults and of Freiburg University for the testing done with the children. Parents of all children gave written informed consent at the beginning of the test session.

Stimuli

Recordings of the emotional sounds were taken from the Montreal Affective Voices database (Belin, Fillion-Bilodeau, & Gosselin, 2008), in which actors were instructed to produce emotional interjections using the vowel *a*. The voices from four identities (two female) expressed anger and fear. Recordings of the phonemes (/m/ and /o/) were made in a double-walled sound attenuated booth at Bangor University using, again, four different German native speakers (two female). The phoneme recordings were edited using CoolEditPro 2.0 (Syntrillium, Inc.) to single utterances of *m* and *o* sounds (i.e., editing was comparable with that of the emotional expressions in Belin et al., 2008). All sounds were normalized in energy (root mean square).

We created morph continua separately for each identity in the emotion and in the phoneme condition using Tandem STRAIGHT (Banno et al., 2007; Kawahara et al., 2008; see also http://www.wakayama-u.ac.jp/~kawahara/STRAIGHTadv/index_e.html for more information). Each continuum consisted of five steps corresponding to 5%/95%, 35%/65%, 50%/50%, 65%/35%, and 95%/5% fear/anger or *m*/*o*. The original angry (0%/100%) and fearful (100%/0%) voices were used as adaptors in the emotion task, and the original *m* (0%/100%) and *o* (100%/0%) sounds were used as adaptors in the phoneme task. Tandem STRAIGHT performs an instantaneous pitch-adaptive spectral smoothing of each stimulus for separation of contributions to the voice signal arising from the glottal source (including F_0) versus supralaryngeal filtering (distribution of spectral peaks, including the first formant frequency, F1). Voice stimuli were decomposed by Tandem STRAIGHT into five parameters: F_0 (the perceived pitch of the voice), frequency, duration, spectrotemporal density, and aperiodicity. Each parameter can be manipulated independently. For each voice, we manually identified one time landmark with three frequency landmarks (corresponding to the first three formants) at the onset of phonation and the same number of landmarks at the offset of phonation. Morphed stimuli were then generated by resynthesis on the basis of the interpolation (linear for time; logarithmic for F_0, frequency, and amplitude) of these time-frequency landmark templates (see also Schweinberger, Kawahara, Simpson, Skuk, & Zäske, 2014, for a recent discussion of the voice morphing technique). Stimuli were normalized in energy (root mean square) before and after morphing. Acoustic analyses to illustrate the validity of the morphing technique with affective bursts, along with sample stimuli, are published elsewhere (Bestelmeyer, Rouger, et al., 2010). The duration of the vocalizations within each continuum was kept constant and ranged between 0.6 and 0.9 s across continua. We used E-Prime 2.0 (Psychology Software Tools, 2012) for stimulus presentation. Sounds were presented via headphones at a comfortable sound level for the child.

To ensure that stimulus categorizations for both emotion and phoneme tasks would yield similar sigmoid-shaped curves with comparable points of subjective equality and slope, we first tested both tasks on 57 young, healthy undergraduate students (35 female; mean age = 20.86, *SD* = 5.19). The slopes of the curves for each task were not significantly different, $t(56)$ =

−1.00, $p = .322$. Both tasks resulted in a significant main effect of adaptation—emotion task: $F(2, 112) = 79.261$, $p < .001$; speech task: $F(2, 112) = 147.64$, $p < .001$. The emotion stimuli are the same as the ones used in several other studies for which categorization consistently resulted in sigmoid-shaped curves (Bestelmeyer, Maurage, Rouger, Latinus, & Belin, 2014; Bestelmeyer, Rouger, et al., 2010; Pye & Bestelmeyer, 2015) and robust contrastive aftereffects in adults.

Procedure

Participants were asked to complete a total of five blocks for each task. All tasks required a two-alternative forced choice judgment. The emotion task required a judgment as to whether the ambiguous vocal morph sounded more angry or fearful, and the phoneme task required a judgment as to whether the vocal morph sounded more like the letter *m* or *o*. Each task consisted of one preadaptation or baseline block, which was always administered before the four adaptation blocks of each task. The trials of the baseline block consisted of individual morphs that had to be categorized. Each baseline block comprised three repetitions of each of the five morph steps per identity (2 male, 2 female), leading to a total of 60 trials. Morphs were presented in randomized order but blocked for gender.

The trial structure of the adaptation tasks consisted of one adapting voice played four times in succession and followed by an ambiguous morph (test stimulus) after a silent gap of 1 s. The intertrial interval was response-terminated at a maximum of 10 s before the next trial would commence. During this interval, the participants categorized the test stimulus using one of two keys ("Does the person sound angry or fearful?"/"Does the person say *m* or *o*?"). Each of the four adaptation blocks (2 Emotion × 2 Gender or 2 Phoneme × 2 Gender) and each of the five test morphs per identity were repeated six times, leading to a total of 30 trials per block. Trials were presented in randomized order but blocked for task type and adaptation condition (i.e., both blocks consisting of female and male fear adaptation were presented in succession rather than mixing the emotion adaptation conditions).

To avoid adaptation to task irrelevant factors such as identity-specific or low-level acoustic properties, participants were always tested on a different identity from the one they were adapted to. Half of the participants were adapted to angry and then fearful voices, whereas the remaining participants were presented with the reverse order of blocks. The same protocol was used for the phoneme adaptation task. Task order was also counterbalanced across participants. The raw data for this experiment are published with Mendeley (doi:10.17632/dzcszhwsfp.1).

Results

For each participant, responses were averaged as a function of the five morph steps, adaptation condition, and task. We then fitted the data with a logistic function with four free parameters: the minimum and maximum x and y values at the center of symmetry of the curve (the point of subjective equality [PSE]) and the slope of the curve at the PSE. Excellent fits were obtained for both tasks across all conditions for each participant group (emotion task for the ASD group: mean $R^2 = .926$, $SD = .186$, and the TD group: mean $R^2 = .956$, $SD = .123$; phoneme task for the ASD group: mean $R^2 = .987$, $SD = .027$, and the TD group: mean $R^2 = .992$, $SD = .013$).

Figure 1 illustrates the psychometric curves fitted to the mean data for the participant groups and tasks. Figures 1a and 1d illustrate the mean responses of the emotion and phoneme baseline conditions (no adaptation), respectively. Figures 1b and 1e illustrate the mean responses of the adaptation conditions to emotions and phonemes, respectively. The group PSE is illustrated with a star on all average curves in corresponding colors. The slopes in the baseline tasks are not significantly different, suggesting similar difficulty levels across tasks. Aftereffect sizes do not differ across participant groups. The bar graphs in Figures 1c and 1f show the aftereffect sizes (Condition A–Condition B at the PSE) for the emotion and phoneme tasks, respectively, as well as individual participants' data (black circles).

To ensure that the tasks did not differ in difficulty, we calculated the slope for the baseline curves. Using a mixed-design analysis of variance (ANOVA; 2 Groups × 2 Tasks), we found no significant main effects of task, $F(1, 49) = .503$, $p = .482$, $\eta_p^2 = .010$, or group, $F(1, 49) = .524$, $p = .472$, $\eta_p^2 = .011$, and no significant interaction between group and task, $F(1, 49) = .217$, $p = .643$, $\eta_p^2 = .004$, on the slope of the curves. Interestingly, from Figure 1a, it is obvious that the extreme emotion categories were less accurately categorized in the children with ASD than the typically developing children (one-way ANOVA for the two extreme morph steps)—anger: $F(1, 49) = 5.124$, $p = .028$; and fear: $F(1, 49) = 5.894$, $p = .019$. This pattern was not evident in the control task (see Fig. 1b), both $F(1, 49) < 1.079$, $p > .304$.

To assess any group differences in aftereffect size in each task and adaptation condition (including the baseline condition), we carried out a mixed-design ANOVA (2 Participant Groups × 2 Tasks × 3 Adaptation Conditions) on the PSE abscissa values, which revealed a significant main effect of adaptation condition, $F(2, 98) = 151.821$, $p < .001$, $\eta_p^2 = .756$. Pairwise comparisons revealed significant differences between all adaptation conditions, including the baseline task (all at $p < .0001$). Contrary to our hypotheses, there was no significant main effect of group, $F(1, 49) = .699$, $p = .407$, $\eta_p^2 =$

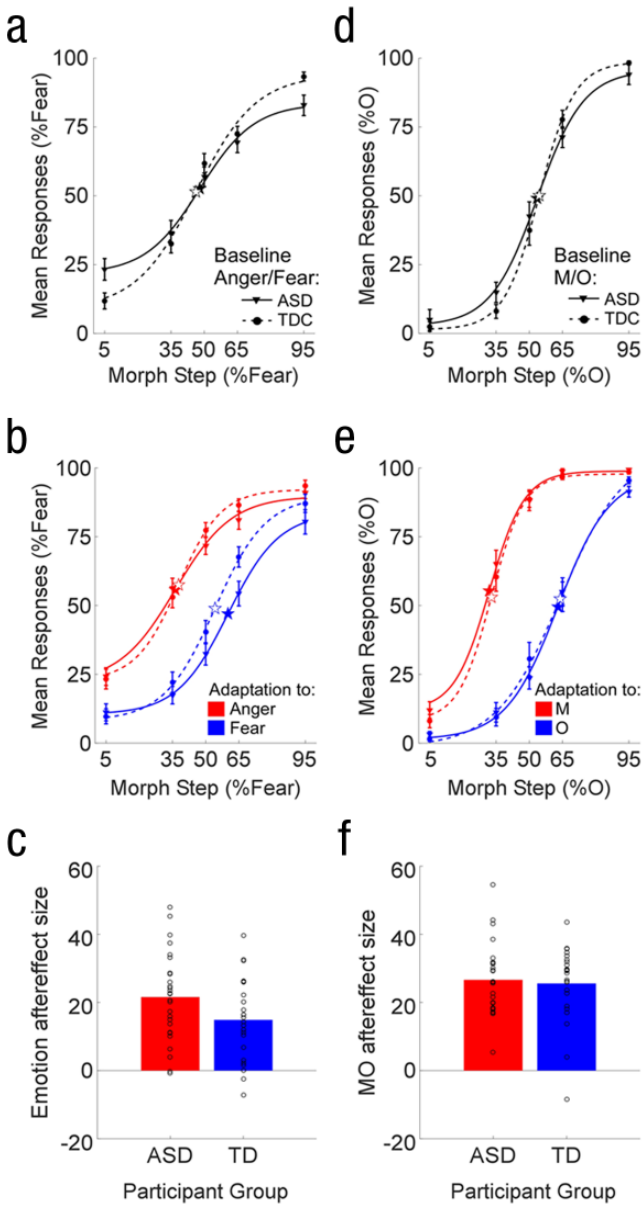

Fig. 1. Illustration of aftereffects for each task, condition, and participant group. Psychometric functions were fitted for the mean responses to each morph step and separately for each task and participant group. Solid lines indicate the performance of the children with autism spectrum disorder (ASD); dashed lines indicate the performance of the typically developing (TD) children. The point of subjective equality (PSE) is plotted as a star on each curve (stars for the PSEs of the children with ASD are filled; stars for the PSEs of the TD children are white). Baseline tasks for the emotion and phoneme tasks are plotted in (a) and (d), respectively. Adaptation tasks for the emotion and phoneme tasks are plotted in (b) and (e). Error bars represent standard error of the mean. The aftereffect sizes for each participant group are based on the PSE in each condition as well as individual participants' data (black circles) are shown in (c) and (f).

.014, and no significant interaction between group and task, $F(1, 49) = .326$, $p = .571$, $\eta_p^2 = .007$. We did find a significant interaction between task and adaptation condition, which was driven by a significantly larger aftereffect to the phoneme sounds than to the vocal expressions. There were no other significant main effects or interactions.

We also examined the possibility that children with ASD adapt more slowly by testing for aftereffect differences between the two groups in only the first half of trials. This analysis resulted in the same pattern as the analysis on the full set of trials (i.e., no main effect of group or significant interaction between group and task). We also explored the possibility that variables such as age, IQ, and autism quotient might covary with our effects of interest, but none of these variables did (all $F < 1.6$, $p > .22$). Similarly, none of these variables significantly correlated with the aftereffect size (PSE of fear–PSE of anger adaptation; PSE of O adaptation–PSE of M adaptation; all $r < -.19$, $p > .19$).

Discussion

The current study tested two key questions: First, will differences in contrastive aftereffects, previously shown using visual stimuli in children with ASD, be evident within the auditory domain? Second, will such differences be seen only in response to the more socially salient stimuli? Recent influential work has suggested that changes in adaptive mechanisms in ASD may be at the root of some of the core aspects of the social phenotype in ASD, including deficits in social cognition and social interaction. However, using auditory stimuli, our within-subjects study revealed no differences in aftereffect sizes for the two participant groups in either the more socially salient or the well-matched control stimuli. In other words, children with ASD did not show any reductions in aftereffect sizes compared with typically developing children regardless of auditory stimulus type. We also found no suggestions of a slowed adaptation rate in children with ASD compared with typically developing children. It is important to note here that this is not simply a null finding; instead, we report robust adaptation effects that are independently significant in each group and virtually identical in magnitude across groups.

Several research groups have reported that the mere categorization or recognition of social stimuli is impaired in ASD (e.g., emotional faces: Harms et al., 2010; Teunisse & de Gelder, 2001; emotional faces and speech prosody: Globerson et al., 2015; Philip et al., 2010; Taylor, Maybery, Grayndler, & Whitehouse, 2015). It is interesting that we observed in the ASD group a pattern similar to the socially salient vocal expressions of anger and fear endpoints in the baseline categorization task, confirming limitations in recognizing emotional expressions also in the auditory domain. In a seminal article, Pellicano et al. (2007) suggested that this poorer performance in categorizing social stimuli may be caused by, or may at least be related to, impairments in the adaptive mechanism in ASD. Our data do

not support this claim. Although we replicated previous findings of poorer categorization performance of the emotional expressions in the ASD group, we found no group differences in the magnitude of the aftereffects to emotional expressions. This pattern of results suggests that impaired recognition of emotions but intact perceptual plasticity in ASD are unlikely to be causally linked.

There were several differences in terms of task design and stimulus type between Pellicano et al.'s (2007) study and the present one, which preclude a definite answer as to whether adaptive coding plays a role in the cognitive phenotype of ASD. Whereas both studies employed social stimuli (facial identity, vocal emotion), our paradigm was not designed around an arguably more complex game. Rather, our study was based on a traditional cognitive adaptation paradigm with four repetitions of the adaptor followed by a test stimulus that is known to yield robust adaptation effects. Although our paradigm may have been less engaging because it was not embedded in a story, it is possible that at least some children, particularly those with ASD, might find our paradigm easier to deal with because no social understanding or involvement was required to complete the task. More specifically, it is possible that the task used by Pellicano et al. was actually less suitable for children with ASD because of the social framing of the task, which might put children with ASD at a disadvantage. Whether the discrepancy in results of the two studies is due to differences in modality or task therefore remains an open question because later face adaptation studies in ASD also typically use tasks designed around a game or story.

Lawson et al. (2015) investigated two types of low-level auditory adaptation in adults with ASD. First, participants had to repeatedly judge the loudness of a quiet, continuously and monaurally presented sound. This type of adaptation (simple loudness) was impaired in adults with ASD compared with healthy controls and may give rise to increased sound sensitivity in ASD. Second, participants had to judge the volume of a much louder, continuously presented sound while a distracter sound was intermittently played to the other ear. This type of adaptation (induced loudness) was intact in the autism group. The authors point out that different neural substrates likely underpin each adaptation type, but it is unclear why the participants with ASD were impaired for simple but not induced loudness adaptation. The results do suggest, however, that the basic adaptation mechanism is not impaired in ASD.

Studies following Pellicano's seminal article have not been entirely unequivocal regarding reduced aftereffects to social stimuli in ASD. Most studies have shown diminished aftereffect sizes (Ewing et al., 2013; Pellicano et al., 2007; Pellicano, Rhodes, & Calder, 2013; Turi et al., 2015; Turi et al., 2016), one recent study has shown a priming effect in the ASD group but not the control group (Rutherford, Troubridge, & Walsh, 2012), and only a few studies, on adult participants with ASD, have found no group differences in the face adaptation effect size (Cook et al., 2014; Walsh, Maurer, et al., 2015; Walsh, Vida, et al., 2015). Participant numbers (or statistical power) are unlikely to explain the differences between previous results and results of the current study, as participant numbers in the current study are larger than in many of the aforementioned reports (all but Ewing et al., 2013, and Walsh, Maurer, et al., 2015). There are several possible explanations for these discrepancies. For example, although adaptation is a ubiquitous and fundamental mechanism across sensory modalities, it is possible that differences or developmental delays in *visual* adaptation are modality specific and do not affect adaptation to high-level auditory stimuli. Equally, it is plausible that auditory stimuli are less prone to attentional lapses than visual stimuli and that, despite attempts to control attention in visual paradigms (Ewing et al., 2013), reduced attention to the adapter stimuli in ASD does play a role, particularly in the child population. It is important that if atypical adaptation to faces is seen only in *children* with ASD, then the effect may be due to delayed or atypical development and therefore would not be a stable aspect of the condition. In addition, as our study demonstrates typical adaptation aftereffects to a "social" (albeit auditory) stimulus in childhood, atypical adaptation in ASD also cannot be assumed in response to all social stimuli. Thus, the suitability of impaired adaptation as an endophenotype is unlikely. At this point, smaller aftereffects demonstrated in some studies but not others may be due to a myriad of factors, but it cannot be seen as a robust or universal impairment. We can therefore exclude impaired aftereffects as reliably descriptive of the cognitive phenotype of ASD.

Our results highlight the need for a more systematic investigation of adaptation in autism to account for the inconsistent findings in the literature. To understand the nature of the putative impairment of the adaptation mechanism in ASD, a thorough investigation of contrastive aftereffects using different stimulus types (social vs. nonsocial) at different levels of the processing hierarchy (low level vs. high level) for various modalities and across different stages of development is required.

Conclusion

We show for the first time that auditory contrastive aftereffects in children with ASD are intact for both socially salient stimuli and control stimuli, thereby

challenging the widely held belief that adaptation mechanisms are specifically impaired in autism spectrum disorders. Whereas children with ASD were significantly worse at categorizing the endpoints of the vocal emotion continua during the baseline task, our results demonstrate that these differences are not directly related to the functioning of the adaptation mechanism. When using a traditional adaptation paradigm with a categorization task on the high-level dimensions of sound, the adaptation mechanism in ASD is intact.

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Declaration of Conflicting Interests

The author(s) declared that there were no conflicts of interest with respect to the authorship or the publication of this article.

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